[INCH–POUND]
A–A–59590/7B
22 April 2010
SUPERSEDING
A–A–59590/7A
17 August 2005

COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

HOLDER, ELECTRICAL CARD, WEDGE RETAINERS, 3 PIECE, SCREW ACTUATED DRIVE, .220 X .220 INCH BODY SIZE

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

The complete requirements for procuring electrical card holders described herein shall consist of this document and the issue in effect of A–A–59590.

CLASSIFICATION/PART IDENTIFICATION NUMBER (PIN). This commercial item description (CID) specification sheet uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see NOTES).

AA59590	/07	B, C, H, R	*	* *	W, D, E
CID number	Applicable CID specification sheet	Finish	Mounting	Length (see table I)	Configuration

Example: AA59590/07HJ48D is the PIN for a hard black anodize finished, 4.8 inch (121.9 mm) long card holder. The card holder also features two counterbored through holes for rivet mounting, and a combination of a lockwasher, flat washer, and screw self-lock element for added resistance to loosening.

SALIENT CHARACTERISTICS.

Performance. Card holders shall hold the circuit card assembly firmly in its installed position. When the card holder is installed properly, it is capable of withstanding 60g/6ms of shock, 25 G-rms of vibration, and provides from 2 to 4 degrees C/W/inch thermal resistance transfer between the circuit card assembly and the heat sink surfaces.

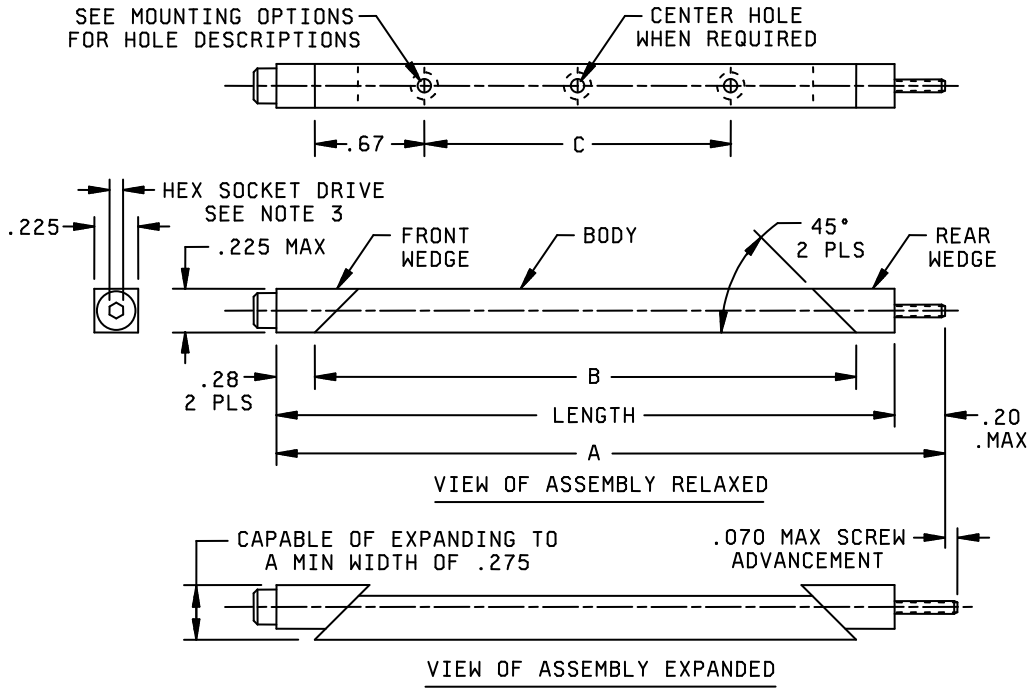
Interface and physical dimensions. The card holders supplied to this CID specification sheet shall be as specified on figures 1, 2, 3, in table I, and A–A–59590.

Actuating screw hex drive socket. The across flats dimension for the hex drive socket shall be as follows: .094 inch (2.38 mm) for mounting options "J", "N", "T", "R", "V", and "S" and .098 inch (2.5 mm) for mounting options "G" and "M".

Material. Unless otherwise specified herein, the card holder materials shall be as specified in A–A–59590.

Nominal installation torque. When card holders are used in cold plate applications, the nominal installation torque of each card holder shall be as follows: 6 inch-pounds (0.68 N-m) for assemblies using no options or option "W" and 8 inch-pounds (0.90 N-m) for assemblies using options "E" or "D".

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Inches	mm	Inches	mm	Inches	mm
.070	1.78	.225	5.72	.28	7.1
.20	5.1	.275	6.99	.67	17.0

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Unless otherwise specified, tolerances are ±.02 inch (0.51 mm) for two place decimals and ±.010 inch (0.25 mm) for three place decimals.
3. The across flats dimension for hex drive socket shall be .094 inch (2.38 mm) for mounting options "J", "N", "T", "R", "V", and "S". The across flats dimension for hex drive socket shall be a metric dimension of .098 inch (2.5 mm) across flats for mounting options "G" and "M".

FIGURE 1. Relaxed and expanded dimensions.

Cold plate slot width dimensions (when applicable). The recommend cold plate slot width to accommodate the circuit card assembly with attached card holder is .250 inch (6.35 mm) plus the thickness of the printed board of the circuit card assembly (see A–A–59590).

Finish. The finish designator shall be as specified in A–A–59590. The finishes available for this CID specification sheet are as follows: "B" (black anodize), "C" (gold chemical film), "H" (hard black anodize), or "R" (clear chemical film).

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Mounting. The mounting designators shall be as specified in A–A–59590. The mounting options available for this CID specification sheet are as follows: "J" (rivet mounting holes), "N" (no mounting holes), "T" or "R" (tapped 0-80 UNF 2B holes), "V" or "S" (tapped 2-56 UNC 2B holes), or "G" or "M" (tapped M2.5 x.45 metric holes). Card holders using mounting option "J" may be shipped unassembled. See figure 1 for mounting hole spacing requirements.

Mounting holes and locations (when required). Two mounting holes are required on card holders less than 5.30 inches (134.6 mm) in length. Three mounting holes are required on card holders 5.30 inches (134.6 mm) or greater in length. The third mounting hole shall be centered on the mounting body. See figure 1 and table I for mounting hole locations and spacing requirements.

Rivet mounting holes. The holes used for rivet mounting shall be .066 to .073 inch (1.68 to 1.85 mm) diameter through holes, countersunk 100 degrees by .140 inch (3.56 mm) diameter with an access/clearance counterbore hole of .156 inch (3.96 mm) diameter by .160 to .175 inch (4.06 to 4.45 mm) deep.

Rivets. This card holder uses rivet type A as specified in A–A–59590 when rivet mounting is used.

Length, expanded, and relaxed dimensions. The length, expanded, and relaxed dimensions shall be as specified on figure 1. The length designator shall be as specified in A–A–59590 and the available lengths for this specification sheet are listed in table I.

TABLE I. Additional assembly dimensions (see figure 1). 1/

PIN length designator	Dimension "LENGTH" ±.02 (0.5)	Dimension "A" (maximum)	Dimension "B" ±.03 (0.8)	Dimension "C" ±.005 (0.13)
18	1.8 (45.7)	2.10 (53.3)	1.24 (31.5)	.75 (19.1)
23	2.3 (58.4)	2.60 (66.0)	1.74 (44.2)	.40 (10.2)
28	2.8 (71.1)	3.10 (78.7)	2.24 (56.8)	.90 (22.9)
30	3.0 (76.2)	3.30 (83.8)	2.44 (62.0)	1.10 (27.9)
33	3.3 (83.8)	3.60 (91.4)	2.74 (69.6)	1.40 (35.6)
35	3.5 (88.9)	3.80 (96.5)	2.94 (74.7)	1.60 (40.6)
38	3.8 (96.5)	4.10 (104.1)	3.24 (82.3)	1.90 (48.3)
43	4.3 (109.2)	4.60 (116.8)	3.74 (95.0)	2.40 (61.0)
48	4.8 (121.9)	5.10 (129.5)	4.24 (107.7)	2.90 (73.7)
53	5.3 (134.6)	5.60 (142.2)	4.74 (120.4)	3.40 (86.4)
58	5.8 (147.3)	6.10 (154.9)	5.24 (133.1)	3.90 (99.1)
63	6.3 (160.0)	6.60 (167.6)	5.74 (145.8)	4.40 (111.8)
68	6.8 (172.7)	7.10 (180.3)	6.24 (158.5)	4.90 (124.5)
73	7.3 (185.4)	7.60 (193.0)	6.74 (171.2)	5.40 (137.2)
78	7.8 (198.1)	8.10 (205.7)	7.24 (183.9)	5.90 (149.9)
83	8.3 (210.8)	8.60 (218.4)	7.74 (196.6)	6.40 (162.6)
88	8.8 (223.5)	9.10 (231.1)	8.24 (209.3)	6.90 (175.3)

1/ Dimensions are in inches. Millimeters, in parenthesis, are given for information only.

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Configuration. The configuration of a card holder shall be as specified in table II. The details of a particular configuration consist of those on figure 1, and may include those on figures 2 and 3. Card holders not requiring the following option shall leave the configuration position in the PIN blank.

TABLE II. Configurations.

Configuration	Applicable figures	Hardware options
	1	No added options
W	1 and 2	Lockwasher and flat washer
E	1 and 3	Screw self-locking element
D	1, 2, and 3	Lockwasher, flat washer and screw self-locking element

Lockwasher and flat washer (see figure 2). The use of a lockwasher and flat washer located under the screw head will provide additional resistance to loosening from shock and vibration. Card holders requiring a lockwasher and flat washer shall include a suffix "W" in the PIN (see classification and notes).

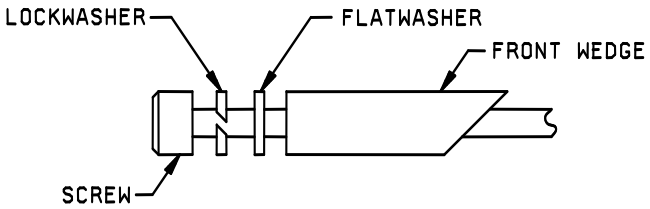


FIGURE 2. Lockwasher and flat washer details.

Screw self-locking element (see figure 3). The use of a screw self-locking element will provide prevailing torque for resistance to loosening from shock and vibration. Card holders requiring a screw self-locking element shall include a suffix "E" in the PIN (see classification and notes).

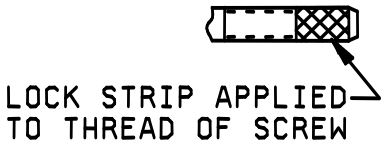


FIGURE 3. Screw self-locking element details.

Lockwasher, flat washer, and screw self-locking element. Card holders requiring a lockwasher, flat washer, and screw self-locking element configuration shall include a suffix "D" in the PIN (see classification and notes).

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NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID specification sheet. See the classification section for PIN format example.

Source of documents.

Commercial Item Description

A–A–59590 – Holder, Electrical Card, Wedge Retainers, 3 Piece, Screw Actuated Drive, General Requirements For.

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or https://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111–5094.)

Ordering data. Ordering data is as specified in A–A–59590.

Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specification sheet preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

Manufacturer CAGE	Manufacturer name and address	Manufacturer contact information
18915	Birtcher – A Division of Pentair Electronic Packaging 7238 Trade Street San Diego, CA 92121–3410	Telephone: (858) 740–2400 Toll Free: (800) 854–7086 Facsimile: (858) 679–4555 URL: www.birtcherproducts.com
52094	Calmark – A Division of Pentair Electronic Packaging 7238 Trade Street San Diego, CA 92121–3410	Telephone: (858) 740–2400 Toll Free: (800) 854–7086 Facsimile: (858) 740–2430 Electronic mail: sales@calmark.com URL: www.calmark.com
5BG68	Card Locks Unlimited, Inc. 2310 E. Orangethorpe Avenue Anaheim, CA 92806–1231	Telephone: (714) 738–6194 Facsimile: (714) 446–0119 E-mail: sales@clumfg.com URL: www.clumfg.com

Part number supersession data. This CID specification sheet PINs supersede the following manufacturer's part numbers in table III and on DSCC Drawing 84103 (table IV) as shown. The CID PINs listed in table III are only for length designator "28". See table V for CID PIN construction using other available lengths for this specification sheet. This information is being provided to assist in reducing proliferation in the Government inventory system.

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TABLE III. Commercial part number supersession data.

PIN designator AA59590/07	Vendor similar designator or type part number 1/ CAGE 18915	Vendor similar designator or type part number 1/ CAGE 52094	Vendor similar designator or type part number 1/ CAGE 5BG68
BJ28	42–6–B	A230–2.80H	3230BA–2.80H
BN28	42–6–B–A	A230–2.80	3230BA–2.80
BT28	42–6–B–T	A230–2.80T0	3230BA–2.80T0
BV28	42–6–B–S	A230–2.80T2	3230BA–2.80T2
BG28	42–6–B–M	MA230–2.80TM2.5	3230MBA–2.80TM2.5
BJ28E	42–6–B–L	A230–2.80HL	3230BA–2.80HL
BN28E	42–6–B–A–L	A230–2.80L	3230BA–2.80L
BT28E	42–6–B–T–L	A230–2.80T0L	3230BA–2.80T0L
BV28E	42–6–B–S–L	A230–2.80T2L	3230BA–2.80T2L
BG28E	42–6–B–M–L	MA230–2.80TM2.5L	3230MBA–2.80TM2.5L
BJ28W	42–6–B–LF	WA230–2.80H	3230WBA–2.80H
BN28W	42–6–B–LF–A	WA230–2.80	3230WBA–2.80
BT28W	42–6–B–LF–T	WA230–2.80T0	3230WBA–2.80T0
BV28W	42–6–B–LF–S	WA230–2.80T2	3230WBA–2.80T2
BG28W	42–6–B–LF–M	MWA230–2.80TM2.5	3230MWBA–2.80TM2.5
BJ28D	42–6–B–LF–L	WA230–2.80HL	3230WBA–2.80HL
BN28D	42–6–B–LF–A–L	WA230–2.80L	3230WBA–2.80L
BT28D	42–6–B–LF–T–L	WA230–2.80T0L	3230WBA–2.80T0L
BV28D	42–6–B–LF–S–L	WA230–2.80T2L	3230WBA–2.80T2L
BG28D	42–6–B–LF–M–L	MWA230–2.80TM2.5L	3230MWBA–2.80TM2.5L
CJ28	42–6	230–2.80H	3230CG–2.80H
CN28	42–6–A	230–2.80	3230CG–2.80
CT28	42–6–T	230–2.80T0	3230CG–2.80T0
CV28	42–6–S	230–2.80T2	3230CG–2.80T2
CG28	42–6–M	M230–2.80TM2.5	3230MCG–2.80TM2.5
CJ28E	42–6–L	230–2.80HL	3230CG–2.80HL
CN28E	42–6–A–L	230–2.80L	3230CG–2.80L
CT28E	42–6–T–L	230–2.80T0L	3230CG–2.80T0L
CV28E	42–6–S–L	230–2.80T2L	3230CG–2.80T2L
CG28E	42–6–M–L	M230–2.80TM2.5L	3230MCG–2.80TM2.5L

See footnote at end of table.

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TABLE III. Commercial part number supersession data – Continued.

PIN designator AA59590/07	Vendor similar designator or type part number 1/ CAGE 18915	Vendor similar designator or type part number 1/ CAGE 52094	Vendor similar designator or type part number 1/ CAGE 5BG68
CJ28W	42–6–LF	W230–2.80H	3230WCG–2.80H
CN28W	42–6–LF–A	W230–2.80	3230WCG–2.80
CT28W	42–6–LF–T	W230–2.80T0	3230WCG–2.80T0
CV28W	42–6–LF–S	W230–2.80T2	3230WCG–2.80T2
CG28W	42–6–LF–M	MW230–2.80TM2.5	3230MWCG–2.80TM2.5
CJ28D	42–6–LF–L	W230–2.80HL	3230WCG–2.80HL
CN28D	42–6–LF–A–L	W230–2.80L	3230WCG–2.80L
CT28D	42–6–LF–T–L	W230–2.80T0L	3230WCG–2.80T0L
CV28D	42–6–LF–S–L	W230–2.80T2L	3230WCG–2.80T2L
CG28D	42–6–LF–M–L	MW230–2.80TM2.5L	3230MWCG–2.80TM2.5L
HJ28	42–6–B3	HA230–2.80H	3230BH–2.80H
HN28	42–6–B3–A	HA230–2.80	3230BH–2.80
HT28	42–6–B3–T	HA230–2.80T0	3230BH–2.80T0
HV28	42–6–B3–S	HA230–2.80T2	3230BH–2.80T2
HG28	42–6–B3–M	MHA230–2.80TM2.5	3230MBH–2.80TM2.5
HJ28E	42–6–B3–L	HA230–2.80HL	3230BH–2.80HL
HN28E	42–6–B3–A–L	HA230–2.80L	3230BH–2.80L
HT28E	42–6–B3–T–L	HA230–2.80T0L	3230BH–2.80T0L
HV28E	42–6–B3–S–L	HA230–2.80T2L	3230BH–2.80T2L
HG28E	42–6–B3–M–L	MHA230–2.80TM2.5L	3230MBH–2.80TM2.5L
HJ28W	42–6–B3–LF	WHA230–2.80H	3230WBH–2.80H
HN28W	42–6–B3–LF–A	WHA230–2.80	3230WBH–2.80
HT28W	42–6–B3–LF–T	WHA230–2.80T0	3230WBH–2.80T0
HV28W	42–6–B3–LF–S	WHA230–2.80T2	3230WBH–2.80T2
HG28W	42–6–B3–LF–M	MWHA230–2.80TM2.5	3230MWBH–2.80TM2.5
HJ28D	42–6–B3–LF–L	WHA230–2.80HL	3230WBH–2.80HL
HN28D	42–6–B3–LF–A–L	WHA230–2.80L	3230WBH–2.80L
HT28D	42–6–B3–LF–T–L	WHA230–2.80T0L	3230WBH–2.80T0L
HV28D	42–6–B3–LF–S–L	WHA230–2.80T2L	3230WBH–2.80T2L
HG28D	42–6–B3–LF–M–L	MWHA230–2.80TM2.5L	3230MWBH–2.80TM2.5L

See footnote at end of table.

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TABLE III. Commercial part number supersession data – Continued.

PIN designator AA59590/07	Vendor similar designator or type part number 1/ CAGE 18915	Vendor similar designator or type part number 1/ CAGE 52094	Vendor similar designator or type part number 1/ CAGE 5BG68
RJ28	42–6–CC	R230–2.80H	3230CC–2.80H
RN28	42–6–CC–A	R230–2.80	3230CC–2.80
RT28	42–6–CC–T	R230–2.80T0	3230CC–2.80T0
RV28	42–6–CC–S	R230–2.80T2	3230CC–2.80T2
RG28	42–6–CC–M	MR230–2.80TM2.5	3230MCC–2.80TM2.5
RJ28E	42–6–CC–L	R230–2.80HL	3230CC–2.80HL
RN28E	42–6–CC–A–L	R230–2.80L	3230CC–2.80L
RT28E	42–6–CC–T–L	R230–2.80T0L	3230CC–2.80T0L
RV28E	42–6–CC–S–L	R230–2.80T2L	3230CC–2.80T2L
RG28E	42–6–CC–M–L	MR230–2.80TM2.5L	3230MCC–2.80TM2.5L
RJ28W	42–6–CC–LF	WR230–2.80H	3230WCC–2.80H
RN28W	42–6–CC–LF–A	WR230–2.80	3230WCC–2.80
RT28W	42–6–CC–LF–T	WR230–2.80T0	3230WCC–2.80T0
RV28W	42–6–CC–LF–S	WR230–2.80T2	3230WCC–2.80T2
RG28W	42–6–CC–LF–M	MRW230–2.80TM2.5	3230MWCC–2.80TM2.5
RJ28D	42–6–CC–LF–L	WR230–2.80HL	3230WCC–2.80HL
RN28D	42–6–CC–LF–A–L	WR230–2.80L	3230WCC–2.80L
RT28D	42–6–CC–LF–T–L	WR230–2.80T0L	3230WCC–2.80T0L
RV28D	42–6–CC–LF–S–L	WR230–2.80T2L	3230WCC–2.80T2L
RG28D	42–6–CC–LF–M–L	MWR230–2.80TM2.5L	3230MWCC–2.80TM2.5L

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.

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TABLE IV. Supersession information.

Superseded (old) PIN in accordance with DSCC Drawing 84103	Superseding (new) CID PIN	Superseded (old) PIN in accordance with DSCC Drawing 84103	Superseding (new) CID PIN
84103–38B	AA59590/07BJ28	84103–38C	AA59590/07CJ28
84103–39B	AA59590/07BJ30	84103–39C	AA59590/07CJ30
84103–40B	AA59590/07BJ35	84103–40C	AA59590/07CJ35
84103–41B	AA59590/07BJ38	84103–41C	AA59590/07CJ38
84103–42B	AA59590/07BJ48	84103–42C	AA59590/07CJ48
84103–43B	AA59590/07BJ53	84103–43C	AA59590/07CJ53
84103–44B	AA59590/07BJ58	84103–44C	AA59590/07CJ58

TABLE V. Example of PIN with available length designators.

PIN designator AA59590/07	Vendor similar designator or type part number 1/ 2/ CAGE 18915	Vendor similar designator or type part number 1/ 2/ CAGE 52094	Vendor similar designator or type part number 1/ 2/ CAGE 5BG68
HJ18D	42–4–B3–LF–L		3230WBH–1.80HL
HJ23D	42–5–B3–LF–L	WHA230–2.30HL	3230WBH–2.30HL
HJ28D	42–6–B3–LF–L	WHA230–2.80HL	3230WBH–2.80HL
HJ30D	3/	WHA230–3.00HL	3230WBH–3.00HL
HJ33D	42–7–B3–LF–L	WHA230–3.30HL	3230WBH–3.30HL
HJ35D	3/	WHA230–3.50HL	3230WBH–3.50HL
HJ38D	42–8–B3–LF–L	WHA230–3.80HL	3230WBH–3.80HL
HJ43D	42–9–B3–LF–L	WHA230–4.30HL	3230WBH–4.30HL
HJ48D	42–10–B3–LF–L	WHA230–4.80HL	3230WBH–4.80HL
HK53D	42–11–B3–LF–L	WHA230–5.30EHL	3230WBH–5.30EHL
HK58D	42–12–B3–LF–L	WHA230–5.80EHL	3230WBH–5.80EHL
HK63D	42–13–B3–LF–L	WHA230–6.30EH	3230WBH–6.30EH
HK68D	42–14–B3–LF–L	WHA230–6.80EHL	3230WBH–6.80EHL
HK73D	42–15–B3–LF–L	WHA230–7.30EHL	3230WBH–7.30EHL
HK78D	42–16–B3–LF–L	WHA230–7.80EHL	3230WBH–7.80EHL
HK83D	42–17–B3–LF–L	WHA230–8.30EHL	3230WBH–8.30EHL
HK88D	42–18–B3–LF–L	WHA230–8.80EHL	3230WBH–8.80EHL

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.
2/ Other lengths are available on request.
3/ Consult manufacturer for part number.

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MILITARY INTERESTS:

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA – CC

Review Activities:
Army – MI
Air Force – 99

CIVIL AGENCY COORDINATING ACTIVITY:

GSA – FAS

Preparing Activity:
DLA – CC

Project 5998–2009–054

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:19Z
Check the source to verify that this is the current version before use.